RAPHAEL SULZER



CONTACT

raphael.sulzer.1@gmail.com
raphaelsulzer.de
Basel, Switzerland

LinkedIn
GitHub
Google Scholar
ResearchGate

SKILLS

Programming

Python	*9 yrs*
C++	*8 yrs*
Bash	*5 yrs*
HTML/CSS	*4 yrs*
C#	*2 yrs*
JavaScript	*2 yrs*

Languages

German	*Native*
English	*Fluent*
French	*Conversational*
Dutch	*Beginner*

Other

3D modeling · aws · Blender · boost · CGAL · claude · CMake · computer vision · conda · Copilot · deep learning · Docker · GIS · geometry processing · GitHub · LaTeX · LiDAR · machine learning · matplotlib · mesh · nanobind · numpy · photogrammetry · point cloud · pytorch · unix · vscode

EXPERIENCE

POSTDOCTORAL RESEARCHER

08/2026 - Present
DIGITAL HUMANITIES LAB, UNIVERSITY OF BASEL, BASEL, SWITZERLAND
Developing an end-to-end pipeline for 3D digitization of cultural heritage objects.

RESEARCH ENGINEER

11/2023 - 07/2026
LUXCARTA TECHNOLOGY, MOUANS-SARTOUX, FRANCE
Developed building reconstruction and vectorization methods combining deep learning and geometry processing on LiDAR point clouds, satellite and aerial imagery. Key contribution: a LOD2 building reconstruction method used to create the national digital twin of France. 3 publications (CVPR, ISPRS, P&RS).

POSTDOCTORAL RESEARCHER

11/2022 - 10/2023
TITANE, INRIA, SOPHIA-ANTIPOLIS, FRANCE
Designed methods for compact polygonal surface reconstruction from point clouds; 1 publication (ECCV).

PHD RESEARCHER

12/2018 - 10/2022
LASTIG, INSTITUT GÉOGRAPHIQUE NATIONAL, PARIS, FRANCE
IMAGINE, ÉCOLE DES PONTS PARISTECH, MARNE-LA-VALLÉE, FRANCE
Investigated learning-based surface reconstruction from real-world point clouds; 4 publications (SGP, ICPR, TPAMI, PE&RS).

LECTURER

12/2018 - 12/2019
ÉCOLE NATIONALE DES SCIENCES GÉOGRAPHIQUES, PARIS, FRANCE
Designed and implemented e-learning courses in photogrammetry and GIS.

GIS DEVELOPER

05/2018 - 07/2019
GEO-COL GIS AND COLLABORATIVE PLANNING, AMSTERDAM, NETHERLANDS
Built custom GIS applications for public and private customers.

GRADUATE STUDENT INTERN

06/2017 - 02/2018
ARUP, AMSTERDAM, NETHERLANDS
Developed an algorithm for building classification from remote sensing and cadastral data; 1 publication (ISPRS).

EDUCATION

PHD DEGREE, DEEP LEARNING AND GEOMETRY PROCESSING

12/2018 - 10/2022
GUSTAVE EIFFEL UNIVERSITY, MARNE-LA-VALLÉE, FRANCE
PhD Thesis: Learning Surface Reconstruction from Point Clouds in the Wild (link)

MASTER'S DEGREE, MSC GEOMATICS, CUM LAUDE

08/2015 - 05/2018
DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS
Master's Thesis: Shape Based Classification of Seismic Building Structural Types (link)

BACHELOR'S DEGREE, BSC GEODESY AND GEOINFORMATICS

10/2011 - 05/2015
UNIVERSITY OF STUTTGART, STUTTGART, GERMANY
Bachelor's Thesis: Photogrammetric Measurement of Snow Depth Using an UAV Platform (link)

SELECTED PUBLICATIONS

Pixels, points and polygons: A dataset and benchmark for multimodal building vectorization

R. Sulzer, L. Duan, N. Girard, F. Lafarge

2026 ISPRS Journal of Photogrammetry and Remote Sensing arXiv

The P^3 Dataset: Pixels, Points and Polygons for Multimodal Building Vectorization

R. Sulzer, L. Duan, N. Girard, F. Lafarge

2025 ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences arXiv

Concise Plane Arrangements for Low-Poly Surface and Volume Modelling

R. Sulzer, F. Lafarge

2024 European Conference on Computer Vision (ECCV) arXiv

SimpliCity: Reconstructing Buildings with Simple Regularized 3D Models

J.-P. Bauchet, **R. Sulzer**, F. Lafarge, Y. Tarabalka

2024 Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) arXiv

A Survey and Benchmark of Automatic Surface Reconstruction from Point Clouds

R. Sulzer, L. Landrieu, R. Marlet, B. Vallet

2024 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) arXiv

Evaluating Surface Mesh Reconstruction Using Real Data

Y. Marchand, L. Caraffa, **R. Sulzer**, E. Clédat, B. Vallet

2023 Photogrammetric Engineering & Remote Sensing (PE&RS) link

Learning Surface Reconstruction from Point Clouds in the Wild

R. Sulzer

2022 Université Gustave Eiffel link

Deep Surface Reconstruction from Point Clouds with Visibility Information

R. Sulzer, L. Landrieu, A. Boulch, R. Marlet, B. Vallet

2022 26th International Conference on Pattern Recognition (ICPR) arXiv

Scalable Surface Reconstruction with Delaunay-Graph Neural Networks

R. Sulzer, L. Landrieu, R. Marlet, B. Vallet

2021 Computer Graphics Forum (CGF) arXiv

Shape Based Classification of Seismic Building Structural Types

R. Sulzer, P. Nourian, M. Palmieri, J. van Gemert

2018 ISPRS International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences link

Track-id: Activity Determination based on Wi-Fi Monitoring

van der Spek, S., Verbree, E., Quak, W., Groeneveld, I. J. D. G., **Sulzer, R.**, Theocharous, E., Xu, Y.

2016 13th International Conference on Location-Based Services (LBS) link

SERVICES

Journal reviewing

- ACM Transactions on Graphics
- IEEE Transactions on Visualization and Computer Graphics
- ISPRS Journal of Photogrammetry and Remote Sensing
- Computer Graphics Forum

Conference reviewing

- ACM SIGGRAPH
- IEEE Conference on Computer Vision and Pattern Recognition

REFERENCES

Florent LAFARGE

florent.lafarge@inria.fr
TITANE, INRIA
Postdoctoral advisor

Loïc LANDRIEU

loic.landrieu@enpc.fr
IMAGINE, ÉCOLE DES PONTS PARISTECH
Doctoral advisor